# 06 / Decision – full 1-3-9-27 screen

What bounded conclusion or next action does the evidence justify?

**EX05 / LEAP Hand | Source analysis | C1 | P: paper; W: project; R: repository**

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Source analysis | C1: A low-cost anthropomorphic hand supports selected manipulation and robot-learning workflows.

## Generate grounded experience

### G1 / Grounded physical problem

G1.1 Mechanism: Is the claimed mechanism explicit, physically credible and open to challenge?

G1.2 Closed loop: Are the implemented body, observations, controller, actions, environment and human roles identified?

G1.3 Task boundary: Are task conditions, episode boundaries, outcomes and exclusions stated?

### G2 / Measurable embodiment

G2.1 Embodiment: Is the embodiment or physical coupling realized and identified at the claimed stage?

G2.2 Observability: Are measurements and references suitable for the claimed outcomes, with relevant validity limits?

G2.3 Interfaces: Are consequential interfaces, action meanings, units and timing described or verified?

### G3 / Controlled physical evidence

G3.1 Commissioning: What limits, resets, interventions, fault handling and recovery are described or tested?

G3.2 Experience record: What observations, actions, outcomes and contextual records are available or described?

G3.3 Grounding gap: How are model, measurement or interpretation connected to physical evidence under comparable conditions?

## Use experience for claims

### U1 / Bounded claim logic

U1.1 Claim: Can the available evidence support or challenge the specific claim, without expanding its scope?

U1.2 Role of experience: Is the role of experience in learning, inference, adaptation, evaluation or design insight explicit?

U1.3 Comparators: Are comparators or alternative explanations appropriate to the contribution being claimed?

### U2 / Representative experience

U2.1 Correspondence: Are data, model, policy, embodiment and evaluation configurations linked where needed?

U2.2 Coverage: Are coverage, sampling, failures, evaluation partitions and leakage addressed for the claim?

U2.3 Governance: What provenance, rights, human participation and data safeguards are reported where relevant?

### U3 / Reproducibility and challenge

U3.1 Reconstruction: What is available to reconstruct the system and evaluation, and what remains unverified?

U3.2 Evaluation: Are protocol, denominators, repetitions, uncertainty and exclusions reported appropriately?

U3.3 Failure analysis: Are failure modes, interventions, competing explanations and negative results examined?

## Manage evidence through lifecycle

### M1 / Evidence traceability

M1.1 Identification: Are claims, configurations, datasets and experiments distinguishable and source-linked?

M1.2 Change control: Are variants, updates or corrections traceable, and is their effect on conclusions assessed?

M1.3 Counterevidence: Are material failures, null findings, ambiguous outcomes and counterevidence retained?

### M2 / Resources, burdens and benefits

M2.1 Resources: Are relevant hardware, compute, time, skills and maintenance resources stated?

M2.2 Data burden: Are collection, annotation, reset, storage and human effort addressed where consequential?

M2.3 Benefit: Is the claimed benefit justified against costs and relevant alternatives within the study?

### M3 / Continuing operation

M3.1 Monitoring: What detects mismatch, degradation, invalid measurements or faults in the claimed setting?

M3.2 Operations: What human authority, maintenance, recovery and update practices are described for this stage?

M3.3 Sustainment: What duration, units, contexts or repeated-use evidence supports the claimed persistence?

**Support (strength for the bounded question)**

0 contradicted; 1 weak; 2 partial; 3 adequate within scope; 4 convergent. NR not reported in inspected scope; NE not examined; NA not applicable with a reason.

**Evidence context (separate from origin, support and access)**

E1 reasoning/design/plan; E2 executed analysis/simulation/digital artifact; E3 physical/integrated experiments, including externally reported results; E4 specified physical coverage/auditability; E5 independent reproduction/sustained operation within scope. Not a universal quality ladder. NR/NE/NA: context normally a dash.

# 06 / Decision – evidence and action record

Source analysis: assess the selected claim; retain unexamined questions.

**EX05 / LEAP Hand | Source analysis | C1 | P: paper; W: project; R: repository**

C1 / LEAP Hand / Public paper and resource scope; no independent reproduction

### G1 / Grounded physical problem

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| G1.1 | P finger mechanism | 3 | E1 |
| G1.2 | R joint-control loop | 3 | E3 |
| G1.3 | P task scope | 3 | E1 |

State: Scoped / Gap: external report only
Reviewer: Check claim scope.

### G2 / Measurable embodiment

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| G2.1 | P realized v1 hand | 3 | E3 |
| G2.2 | P force/task tests | 2 | E3 |
| G2.3 | R motor mapping | 2 | E2 |

State: Scoped / Gap: external report only
Reviewer: Check units/reference.

### G3 / Controlled physical evidence

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| G3.1 | R limits/troubleshooting | 2 | E1 |
| G3.2 | W learning workflows | 2 | E3 |
| G3.3 | P sim/real distinction | 2 | E3 |

State: Scoped / Gap: external report only
Reviewer: Inspect one episode.

### U1 / Bounded claim logic

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| U1.1 | P platform contribution | 3 | E3 |
| U1.2 | P learning applications | 3 | E1 |
| U1.3 | P platform baselines | 3 | E3 |

State: Scoped / Gap: external report only
Reviewer: Compare bounded claims.

### U2 / Representative experience

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| U2.1 | R joint conventions | 2 | E2 |
| U2.2 | NE task coverage audit | NE | - |
| U2.3 | NE rights review | NE | - |

State: Open / Gap: NE entries unreviewed
Reviewer: Inspect split/provenance.

### U3 / Reproducibility and challenge

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| U3.1 | W CAD/API resources | 2 | E2 |
| U3.2 | P task-specific tests | 2 | E3 |
| U3.3 | R faults/limitations | 2 | E3 |

State: Scoped / Gap: external report only
Reviewer: Try one cited route.

### M1 / Evidence traceability

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| M1.1 | W v1 identity | 2 | E2 |
| M1.2 | W v1/v2 separation | NE | - |
| M1.3 | P bounded outcomes | 2 | E3 |

State: Open / Gap: NE entries unreviewed
Reviewer: Pin source versions.

### M2 / Resources, burdens and benefits

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| M2.1 | W assembly/resources | 2 | E1 |
| M2.2 | NE full data burden | NE | - |
| M2.3 | P cost comparison | 2 | E3 |

State: Open / Gap: NE entries unreviewed
Reviewer: Count human/compute cost.

### M3 / Continuing operation

| Check | Evidence ID + basis | Support | Context |
|---|---|---|---|
| M3.1 | R actuator faults | NE | - |
| M3.2 | R operator setup | 2 | E1 |
| M3.3 | P repeatability tests | 2 | E3 |

State: Open / Gap: NE entries unreviewed
Reviewer: Check recovery limits.

## Choose the conclusion for the declared mode

Source analysis. Provisional support: 2 partial; 3 adequate for the bounded question; NE not examined. E1 reasoning/design; E2 digital artifact; E3 reported physical experiment. No mean score or deployment approval.

Project-development action: not assessed here. A published result does not authorize the student team to deploy a robot.

Bounded conclusion: LEAP is a documented manipulation platform; its existence does not prove every learned policy transfers reliably.

Follow-up: Reviewer: select one v1 task and identify the object-state measurement needed for a fair physical comparison.

Reassess if data, calibration, implementation or source conclusions change. Next project action requires its own evidence.

# Worksheet 6 / Decision – evidence and team review

EX05 / LEAP Hand | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX05  Project: LEAP Hand  Claim ID: C1  Revision: draft 1

Mode: Source analysis  Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: paper sections and public documentation; no dataset download or independent experiment.

| Evidence ID | Artifact/source + location (clickable) | What it supports; limitation / access gap |
|---|---|---|
| **P** | arxiv.org/abs/2309.06440 | Paper: method/results; reported evidence, not our measurement. |
| **W** | v1.leaphand.com | Project: system/videos; selected examples do not establish reliability. |
| **R** | github.com/leap-hand/LEAP_Hand_API | Repository: documented route; code and data not executed or audited. |

**Next action / owner / verification:**

Reviewer: select one v1 task and identify the object-state measurement needed for a fair physical comparison.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

2. Name / student ID: _____

Signature: _____ Date: _____

3. Name / student ID: _____

Signature: _____ Date: _____

4. Name / student ID: _____

Signature: _____ Date: _____

5. Name / student ID: _____

Signature: _____ Date: _____

6. Name / student ID: _____

Signature: _____ Date: _____

7. Name / student ID: _____

Signature: _____ Date: _____

8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.